

CSE 309 Theory of Automata, Spring 2026
Exercise 6

1. **Turing Machines and Computers** For each of the following, say whether the statement is true or false.
 - (a) If a decision problem can be solved on a computer, then the language corresponding to the decision problem is decided by some Turing machine.
 - (b) If a decision problem can be solved on a computer, then the language corresponding to the decision problem is accepted by some finite automaton.
 - (c) If a language is accepted by some finite automaton, then the decision problem corresponding to the language can be decided by a computer.
 - (d) If a language is accepted by some finite automaton, there is some Turing machine that also decides the language.
 - (e) If a language is decided by some Turing machine, then the language is generated by some context free grammar.
2. **TM Variants** (Question 3.17, Page 189 from Sipser.) Say that a *write-once Turing machine* is a single-tape TM that can alter each tape square at most once (including the input portion of the tape). Show that this variant Turing machine model is equivalent to the ordinary Turing machine model. (*Hint: As a first step, consider the case whereby the Turing machine may alter each tape square at most twice. Use lots of tape.*)
3. **Halting Problem.** Review the undecidability proof of the halting problem discussed in class. In particular, work through the analysis to determine why we do not obtain a contradiction if the program P is executed on some input Turing machine A such that $A \neq P$. In other words, what happens when P receives $\langle A \rangle$ as input.